

ACKNOWLEDGMENTS

The authors would like to thank Haimanot Belachew, Ulas Berk Karli, Ziyao Shangguan, Anushka Potdar, Jirachaya “Fern” Limprayoon, Rebecca Ramnauth, Nicholas C. Georgiou, Debasmita Ghose, Liam Hoffmeister-Ricke, Sasha Lew and Paul Comeau for their feedback, contributions and support.

A APPENDIX

A.1 Feature Details

A.1.1 Optimal Trajectories

We generate a reference trajectory using a PID controller following the original pre-planned trajectory to obtain the optimal velocities v_t^{opt} and optimal positions x_t^{opt} used to compute the alignment-related features. At each time step t along an original trajectory, the controller produced a “natural” trajectory from the robot’s current state—specifically its position x_t and velocity v_t —toward the goal G^* . We consider this reference trajectory to be “optimal” in the sense that it reflects what a human might intuitively expect the robot to do from that point onward, based on the smoothness and predictability of PID-generated motion [3]. We then compute the position and velocity by interpolating the dynamics 0.25 seconds into the future along this reference trajectory to represent x_t^{opt} and v_t^{opt} , respectively.

A.1.2 Legibility

To represent legibility, we adopt the formulation from Dragan et al. [1] and compute a score l_t at each time step t as:

$$l_t = \frac{\int \mathbb{P}(G^* \mid \xi_{S \rightarrow x_t}) \gamma_t dt}{\int \gamma_t dt}, \quad (1)$$

where $\gamma_t = T - t$ is a discount factor that gives more weight to earlier parts of the trajectory and T is the total trajectory duration. The term $\mathbb{P}(G^* \mid \xi_{S \rightarrow x_t})$ is the probability that the robot is going to goal G^* , given the observed trajectory prefix $\xi_{S \rightarrow x_t}$, and is computed as:

$$\mathbb{P}(G^* \mid \xi_{S \rightarrow x_t}) \propto \frac{\exp(-C(\xi_{S \rightarrow x_t}) - C(\xi_{x_t \rightarrow G^*}^{\text{opt}}))}{\exp(-C(\xi_{S \rightarrow G^*}^{\text{opt}}))} \mathbb{P}(G^*). \quad (2)$$

This probability measures how efficient it is to reach the goal G^* via the current observed trajectory snippet $\xi_{S \rightarrow x_t}$, followed by an optimal completion $\xi_{x_t \rightarrow G^*}^{\text{opt}}$, relative to the cost of the optimal trajectory $\xi_{S \rightarrow G^*}^{\text{opt}}$. In our implementation, we define the cost function $C(\cdot)$ as the total trajectory length. Higher legibility indicates that the robot’s motion strongly suggests its intention to reach the goal G^* early on.

A.2 Derivation of the Goal Inference Formulation

A.2.1 Inferring from Where People Grasp the Gripper

First we derive the posterior over goals given the release position is

$$\mathbb{P}(g \mid c_l) \propto \mathbb{P}(c_l \mid g) \cdot \mathbb{P}(g) \propto \mathbb{P}(c_l \mid g), \quad (3)$$

assuming a uniform prior over goals.

During testing, for each trajectory, given the grasp position c_p and velocity c'_p , the MLP predicts the release position:

$$\hat{c}_l = f_{\text{MLP}}(c_p, c'_p). \quad (4)$$

We assume c_l and $\hat{c}_l$ follow the same distribution. Through the MLP and using the Baye’s Rule from Eq. 3, the posterior over goals conditioned on the grasp position and velocity becomes (defined as **WHERE model** goal distribution):

$$\mathbb{P}(g \mid c_p, c'_p) = \mathbb{P}(g \mid \hat{c}_l) \propto \mathbb{P}(\hat{c}_l \mid g) \approx \mathbb{P}_{\text{GMM}}(\text{MLP}(c_p, c'_p) \mid g), \quad (5)$$

where $\mathbb{P}(\hat{c}_l \mid g)$ is calculated using the GMM fitted for $\mathbb{P}(c_l \mid g)$.

A.2.2 Combining When and Where

To derive the posterior over goals using both spatial and timing information:

$$\mathbb{P}(g \mid t_c, c_p, c'_p, \xi) = \frac{\mathbb{P}(t_c, c_p, c'_p, \xi \mid g) \mathbb{P}(g)}{\mathbb{P}(t_c, c_p, c'_p, \xi)}. \quad (6)$$

We expand the joint probability in the numerator using the chain rule:

$$\mathbb{P}(t_c, c_p, c'_p, \xi \mid g) = \mathbb{P}(t_c, c_p, c'_p \mid g, \xi) \mathbb{P}(\xi \mid g). \quad (7)$$

In our setup, the robot’s pre-planned trajectories are generated independently of the sampled goal hypotheses. Therefore, we assume that the trajectory ξ and the goal g are independent a priori, such that $\mathbb{P}(\xi | g) = \mathbb{P}(\xi)$. Substituting this into Eq. 6 gives:

$$\mathbb{P}(g | t_c, c_p, c'_p, \xi) = \frac{\mathbb{P}(t_c, c_p, c'_p | g, \xi) \mathbb{P}(g)}{\mathbb{P}(t_c, c_p, c'_p | \xi)}. \quad (8)$$

We assume conditional independence between the correction timing t_c and the correction initiation parameters (c_p, c'_p) given the goal g and the trajectory ξ , as they capture distinct aspects of the human intervention process: *when* the human decides to intervene, whereas (c_p, c'_p) describe *how* the correction is initiated spatially and dynamically.:

$$\mathbb{P}(t_c, c_p, c'_p | g, \xi) = \mathbb{P}(t_c | g, \xi) \mathbb{P}(c_p, c'_p | g, \xi). \quad (9)$$

This assumption is supported empirically in our correlation analysis (Appendix A.3.1).

Substituting back:

$$\mathbb{P}(g | t_c, c_p, c'_p, \xi) = \frac{\mathbb{P}(t_c | g, \xi) \mathbb{P}(c_p, c'_p | g, \xi) \mathbb{P}(g)}{\mathbb{P}(t_c, c_p, c'_p | \xi)}. \quad (10)$$

The denominator is a normalization constant computed as the sum over all possible goals $\mathcal{G}$.

$$\mathbb{P}(t_c, c_p, c'_p | \xi) = \sum_{\hat{g} \in \mathcal{G}} \mathbb{P}(t_c | \hat{g}, \xi) \mathbb{P}(c_p, c'_p | \hat{g}, \xi) \mathbb{P}(\hat{g}). \quad (11)$$

Thus, we have

$$\mathbb{P}(g | t_c, c_p, c'_p, \xi) = \frac{\mathbb{P}(t_c | g, \xi) \mathbb{P}(c_p, c'_p | g, \xi) \mathbb{P}(g)}{\sum_{\hat{g} \in \mathcal{G}} \mathbb{P}(t_c | \hat{g}, \xi) \mathbb{P}(c_p, c'_p | \hat{g}, \xi) \mathbb{P}(\hat{g})}. \quad (12)$$

To calculate $\mathbb{P}(c_p, c'_p | \hat{g}, \xi)$, we assume that c_p, c'_p is independent of ξ since the correction initiation should reflect human’s intent based on the task goal rather than the shape of the robot’s pre-planned trajectory, so it becomes $\mathbb{P}(c_p, c'_p | \hat{g})$. This assumption is partially supported by our correlation analysis in Appendix A.3.2, where the grasp position c_p exhibits negligible correlation with trajectory similarity across different ξ ’s. c'_p is dependent on where the correction begins, so the weak dependence of c_p suggests that the influence of ξ on correction initiation can be reasonably ignored for our modeling purposes.

Eventually, the equation becomes:

$$\mathbb{P}(g | t_c, c_p, c'_p, \xi) = \frac{\mathbb{P}(t_c | g, \xi) \mathbb{P}(c_p, c'_p | g) \mathbb{P}(g)}{\sum_{\hat{g} \in \mathcal{G}} \mathbb{P}(t_c | \hat{g}, \xi) \mathbb{P}(c_p, c'_p | \hat{g}) \mathbb{P}(\hat{g})}. \quad (13)$$

Potential goals are uniformly sampled, thus $P(\hat{g})$ is uniform, so the posterior (**COMBINED model** goal distribution posterior) becomes:

$$\mathbb{P}(g | t_c, c_p, c'_p, \xi) = \frac{\mathbb{P}(t_c | g, \xi) \mathbb{P}(c_p, c'_p | g)}{\sum_{\hat{g} \in \mathcal{G}} \mathbb{P}(t_c | \hat{g}, \xi) \mathbb{P}(c_p, c'_p | \hat{g})}, \quad (14)$$

where $\mathbb{P}(t_c | g, \xi)$ is derived from Eq.1 from the main paper, and through the MLP:

$$\mathbb{P}(c_p, c'_p | g) \propto \mathbb{P}(\hat{c}_l | g) = \mathbb{P}_{\text{GMM}}(\text{MLP}(c_p, c'_p) | g). \quad (15)$$

Thus,

$$\mathbb{P}(g | t_c, c_p, c'_p, \xi) = \frac{\mathbb{P}(t_c | g, \xi) \mathbb{P}_{\text{GMM}}(\text{MLP}(c_p, c'_p) | g)}{\sum_{\hat{g} \in \mathcal{G}} \mathbb{P}(t_c | \hat{g}, \xi) \mathbb{P}_{\text{GMM}}(\text{MLP}(c_p, c'_p) | \hat{g})}. \quad (16)$$

A.3 Correlations

A.3.1 Correlation Between Correction Timing t_c and Spatial Information (c_p, c'_p)

To verify the assumption that correction timing is largely independent of spatial displacement, we computed the Pearson correlation between the normalized correction timing (in seconds) and the grasp positions relative to the actual goal and the initial correction velocity along each spatial dimension. Across all 3,589 filtered correction samples, we observe the following correlations for the grasp position c_p and the initial correction velocity c'_p :

The results show weak correlations between t_c and both x and y components of c_p and c'_p , suggesting that the timing of intervention is largely independent of where and how participants initiate the correction in the horizontal plane. The moderate correlations along the z -axis likely reflect the vertical component of the motion, which is peripheral to goal inference that primarily occurs in the x - y plane. Therefore, we approximate t_c and (c_p, c'_p) as conditionally independent given g and ξ .

Table 1: Pearson correlation between normalized correction timing t_c and correction spatial features.

	Corr. w/ x	Corr. w/ y	Corr. w/ z
Grasp position c_p	0.092	0.039	-0.600
Initial correction velocity c'_p	0.077	-0.010	0.264

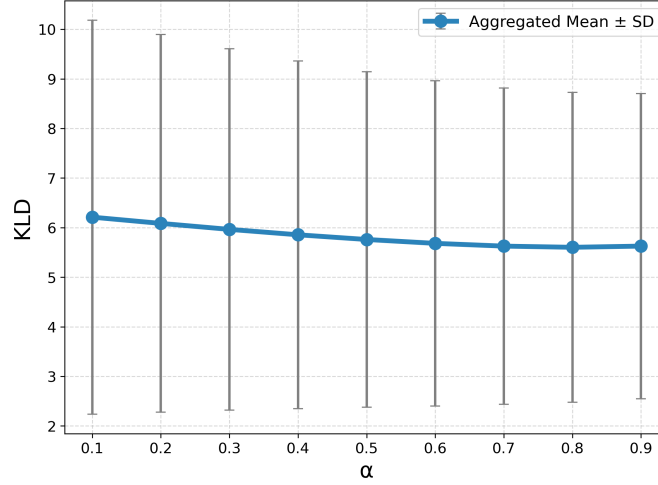


Figure 1: KLD vs. different weight α

A.3.2 Correlation Between Grasp Position c_p and Trajectory ξ Similarities

To verify that the spatial similarity between grasp locations is independent from the overall trajectory similarity, we computed both dynamic time warping (DTW) [4] distances between full correction trajectories and Euclidean distances between their corresponding correction endpoints. From 3,589 correction trajectories, we randomly sampled 353,500 trajectory pairs and calculated the Pearson and Spearman correlations between the two distance measures.

The Pearson correlation coefficient was $r = 0.02$ ($p < 0.001$) and the Spearman rank correlation was $\rho = 0.04$ ($p < 0.001$), both indicating negligible relationships between trajectory and endpoint similarities. These results confirm that trajectories that are temporally or kinematically similar do not necessarily result in spatially similar correction endpoints. Therefore, it is reasonable to treat the temporal and spatial components of human corrections as independent in our modeling framework.

A.4 Model Details

A.4.1 Transformer

We optimize binary cross-entropy over valid time steps only, using the mask M as sample weights. We use Adam [2] with gradient clipping ($\|g\|_2$ clipnorm = 1.0) and an exponentially decaying learning rate ($\eta_0=10^{-3}$, decay_steps = 1000, decay_rate = 0.9). Model selection is done via a held-out validation set by monitoring validation loss and saving the best checkpoint. Random seeds are fixed for reproducibility. Input features are rescaled with the same preprocessing on train/validation splits.

A.4.2 MLP

The model is trained using mean squared error (MSE) loss and optimized with the Adam [2] optimizer with an exponentially decayed learning rate. Validation loss is monitored at each epoch, and the best-performing model is saved based on the lowest validation loss. All inputs and outputs are normalized using statistics computed from the training set to ensure stable convergence.

A.5 Geometric Weight α

To determine the balance between timing and spatial information in the COMBINED model, we varied $\alpha \in [0.1, 0.9]$ in the weighted posterior:

$$\mathbb{P}_w(g \mid t_c, c_p, c'_p, \xi) = \frac{\mathbb{P}(t_c \mid g, \xi)^\alpha \cdot \mathbb{P}_{\text{GMM}}(\text{MLP}(c_p, c'_p) \mid g)^{1-\alpha}}{\sum_{\hat{g} \in \mathcal{G}} \mathbb{P}(t_c \mid \hat{g}, \xi)^\alpha \cdot \mathbb{P}_{\text{GMM}}(\text{MLP}(c_p, c'_p) \mid \hat{g})^{1-\alpha}}. \quad (17)$$

For testing, we evaluated how the KLD between the inferred and ground-truth goal distributions (using correction onset information c_p, c'_p) varies with α at 80% correction timing percentages, aggregated across all targets and shapes over 5 runs. The results in Fig. 1 show that $\alpha = 0.8$ yields the lowest mean KLD. Therefore, we set $\alpha = 0.8$ as the weighting parameter for all COMBINED models.

A.6 KLD Plot for Different Targets Using Correction Onset Information

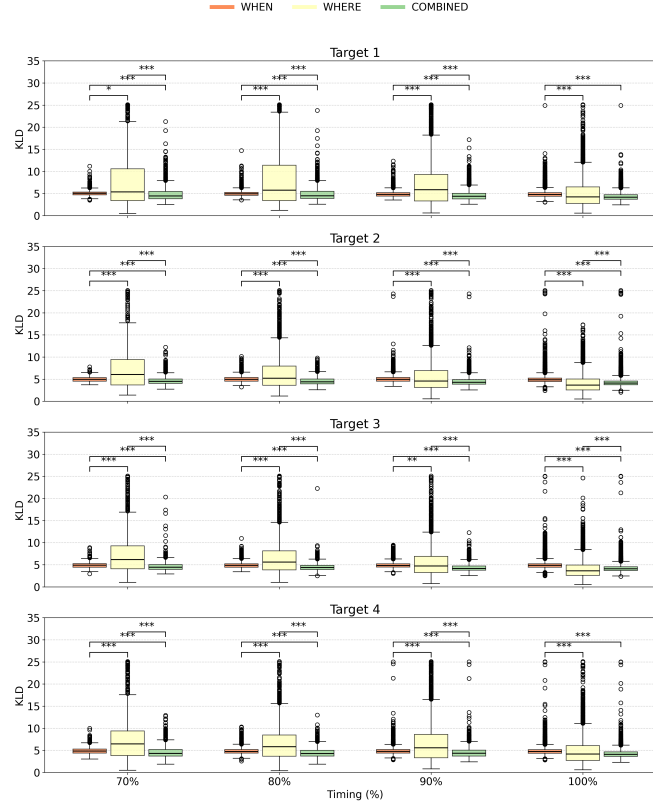


Figure 2: KLD between the ground-truth goal distributions and the inferred goal using the start of the correction information after each correction in the test set, evaluated over 50 random dataset splits. Results are shown for each goal inference model (WHEN, WHERE, and COMBINED) across different correction timing percentages. The results are aggregated across shapes.

We evaluate goal inference performance at each target using the grasp location c_p and initial velocity c'_p as inputs. As shown in Fig. 2, the COMBINED model yields significantly lower KLD than either the WHEN or WHERE models for all targets when earlier corrections are considered ($\leq 70\%$, $\leq 80\%$, $\leq 90\%$). When late corrections are included ($\leq 100\%$), the COMBINED model performs comparably to WHERE for side targets (targets 1 and 4; no significant difference) but slightly worse for center targets (targets 2 and 3). These results suggest that timing information is most beneficial for early corrections and becomes less informative when corrections occur late and target distinctions are less pronounced.

REFERENCES

- [1] Anca D Dragan, Kenton CT Lee, and Siddhartha S Srinivasa. 2013. Legibility and predictability of robot motion. In *2013 8th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 301–308.
- [2] Diederik P Kingma and Jimmy Ba. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* (2014).
- [3] Pauline Maurice, Meghan E Huber, Neville Hogan, and Dagmar Sternad. 2017. Velocity-curvature patterns limit human-robot physical interaction. *IEEE robotics and automation letters* 3, 1 (2017), 249–256.
- [4] Pavel Senin. 2008. Dynamic time warping algorithm review. *Information and Computer Science Department University of Hawaii at Manoa Honolulu, USA* 855, 1–23 (2008), 40.